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How to Release Your Flutter App on the iOS App Store

#dart

#flutter

#production



Andrea Bizzotto

Dec 3, 2024 · 13 min read

Your Flutter app is ready, and you've completed all the important pre-release steps. Now it's time for the final step: publishing it on the **App Store** and sharing it with the world. But where do you start?

Sure, the Flutter documentation provides a technical guide for building and releasing an iOS app. But publishing your first app involves more than just building and uploading—it's a process filled with **red tape**, **guidelines**, and **steps that aren't always obvious**.

If you're new to this, don't worry. This article will provide a **high-level roadmap** for releasing your Flutter app on the App Store. Even if you're familiar with this process, you may discover a trick or two along the way. 🙌

What You'll Learn:

1. How to **enroll in the Apple Developer Program**.
2. How to **register your App ID** in the Apple Developer Portal.
3. How to **create your app** in App Store Connect.
4. How to **prepare your app for review**, including metadata, privacy, and compliance.
5. How to **create a Privacy Manifest** in Xcode.
6. How to **update the Xcode project settings**, including code signing.
7. How to **build, upload, and submit** your app for App Store review.

Want a more detailed, hands-on guide? My [Flutter in Production](#) course includes a complete module with 20 lessons that cover each step in depth—from account setup to submission and beyond.

Let's dive in! 🚀

Should You Sell Apps on the App Store?

For many developers, the **App Store** is the ultimate platform for reaching a high-quality audience. Apple users are far more willing to pay for apps compared to Android users (it's not even close). And with payments seamlessly linked to their Apple ID, purchasing is frictionless.

However, joining Apple's **walled garden** comes with trade-offs:

- **15% revenue cut:** Even with the [small business program](#), Apple takes a significant cut of your earnings.
- **No direct customer ownership:** Apple controls the relationship with your customers, making it harder to connect with them or market additional products.
- **App review delays:** The app review process can take hours or days—and even longer if your app is rejected, causing delays in your release timeline.

The App Store offers real opportunities, but remember that Apple stands **between** you and your users. If your app fits Apple's ecosystem and you're prepared to work within its constraints, the App Store can be a powerful platform to reach millions of users worldwide.

1. Enroll in the Apple Developer Program

To publish apps on the App Store, you'll need to join the [Apple Developer Program](#):

Enroll in the program with the Apple Developer app >

Become a member

To become an Apple Developer Program member, you can enroll as an individual or enroll your organization. Learn about what you'll need and the requirements to enroll.

Start your enrollment

Apple Developer Program: enrollment page

- **Cost:** \$99 per year.
- **Who can enroll:** Individuals or organizations.
- **How to enroll:** Start the process at developer.apple.com/programs/enroll/.
- **Reduce fees:** Join the [Small Business Program](#) to pay a **15% commission** instead of 30% (eligibility applies).

Enrollment Steps

The enrollment process is fairly simple and involves the following steps:

- [Start your Enrollment](#)
- Sign in with your Apple ID (or create an account)
- Agree to the Apple Developer Agreement
- Confirm your personal information
- Select your entity type (individual, organization, or others)
- Obtain a D-U-N-S Number using [this service](#) (organizations only - can take **up to 30 business days**, or up to **8 business days** for the expedited service)
- Agree to the legal terms
- Purchase your membership
- Complete the verification process

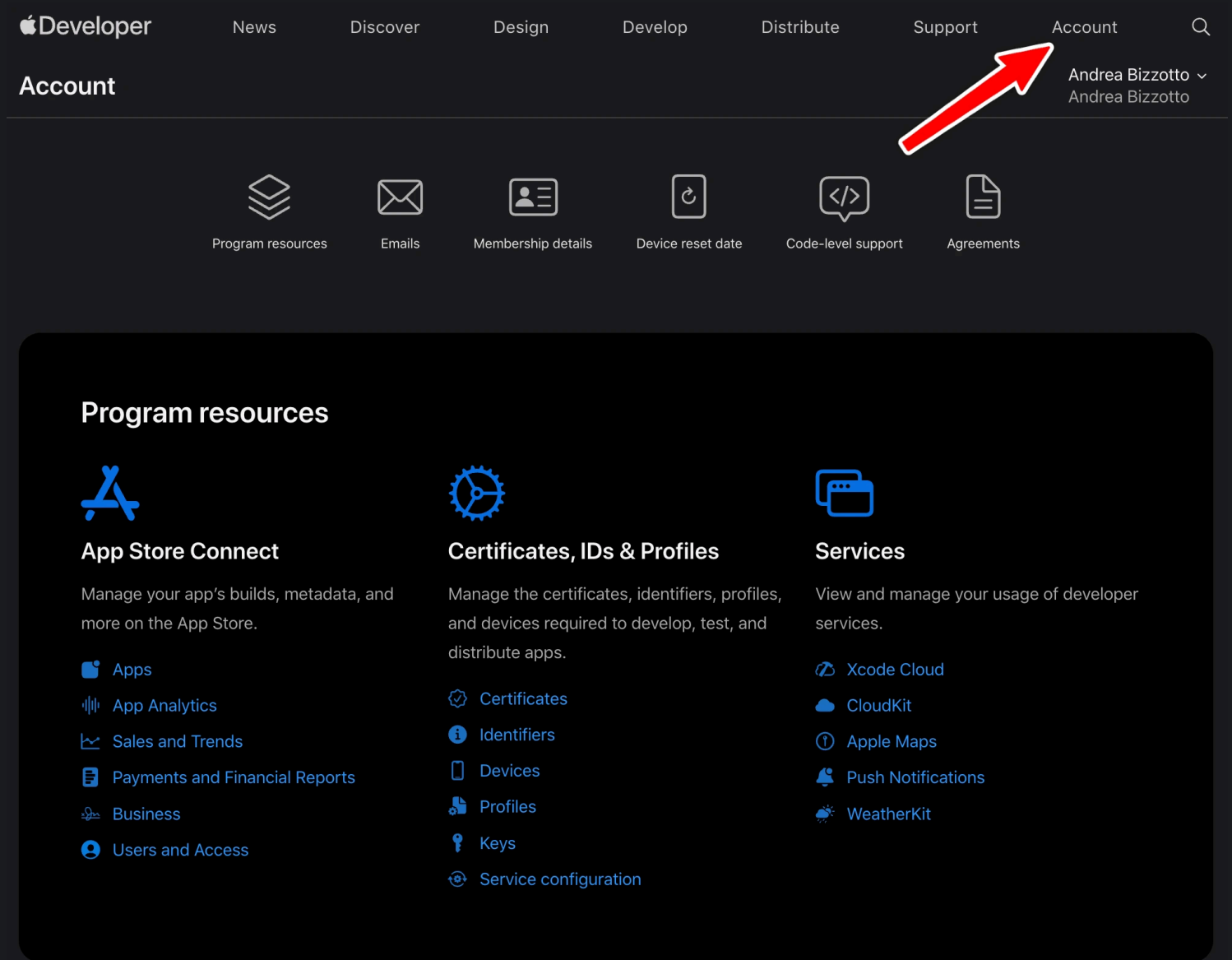
What Happens Next?

Once you've submitted all the required information:

- **Verification:** Apple may take a few days to verify your enrollment.
- **Activation:** Once verified, you'll receive an activation email.
- **Access to Resources:** With your membership, you'll have access to your Apple Developer Account, documentation, and additional resources.

2. Register Your App ID

After enrolling in the Apple Developer Program, you'll get access to your Apple Developer Account:

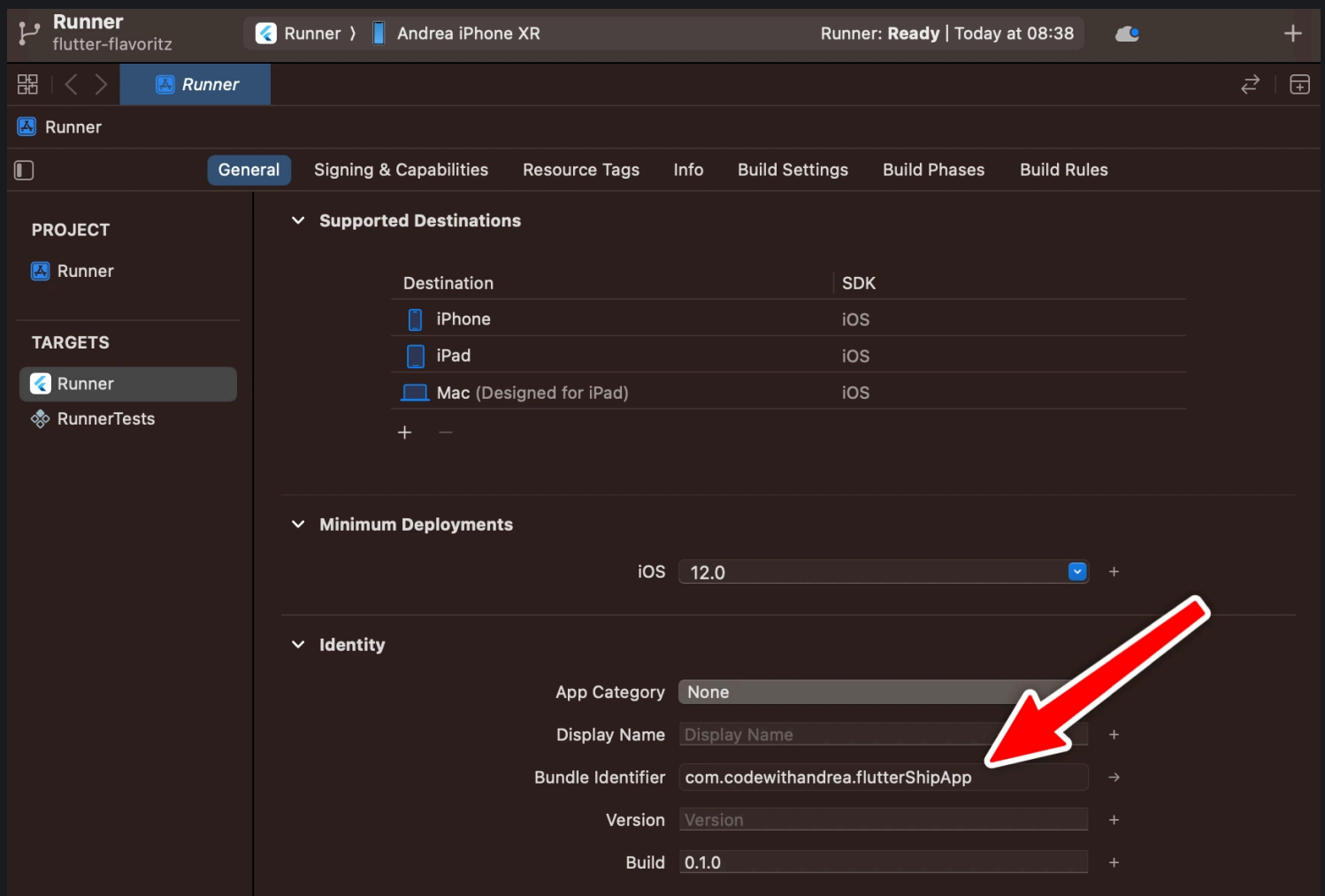


The next step is to **register your App ID**, which uniquely identifies your app within Apple's ecosystem.

How to Register Your App ID

1. Open the identifiers page of your Apple Developer account.
2. Click + to register a new App ID.
3. Enter an app name, select **Explicit App ID**, and enter the Bundle ID (e.g., `com.yourcompany.yourapp`).
4. Select the **capabilities** your app requires (e.g., Push Notifications, In-App Purchases), then click **Continue**.
5. Click **Register** to confirm your App ID.

Note that the **Bundle ID** must match the **Bundle Identifier** property in Xcode under **Runner > General > Identity**:



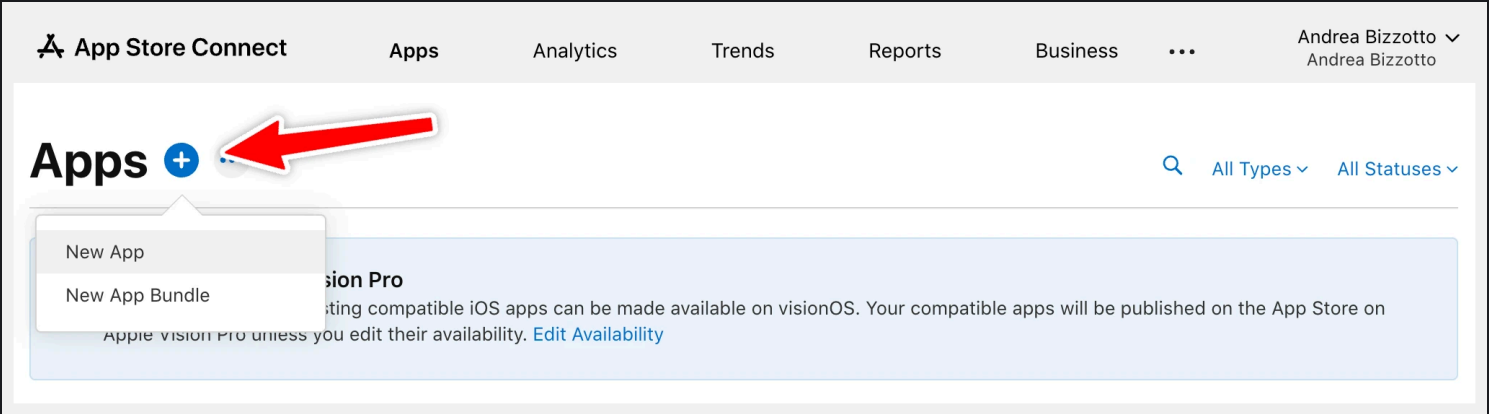
Where to find the Bundle Identifier in Xcode

3. Create your app in App Store Connect

After registering your App ID, the next step is to create your app in [App Store Connect](#).

How to Create a New App in App Store Connect

Go to [App Store Connect > Apps](#), then add a new app:



To create a new app, click the "+" button

Fill in the required details:

New App

Platforms ?

☒ iOS ☐ macOS ☐ tvOS ☐ visionOS

Name ?

Flutter Ship

18

Primary Language ?

English (U.K.)



Bundle ID ?

Flutter Ship Stg - com.codewithandrea.flutterShipApp.stg



SKU ?

com.codewithandrea.flutterShipApp.stg

User Access ?

☐ Limited Access ☒ Full Access

Cancel

Create

Adding the new app details

These include:

- **Platforms:** Ensure **iOS** is selected.
- **Name:** The app name displayed to users on the App Store.
- **Primary Language:** The default language for your app's metadata.
- **Bundle ID:** Select the App ID you registered in the previous step.
- **SKU:** A unique internal ID for your app (not visible to users).
- **User Access:** Decide which App Store Connect users can manage the app.

When you're done, click **Create** to finalize the setup.

4. Prepare your App For Review

After clicking **Create**, you'll land on the app's overview page:

Flutter Ship

Distribution

TestFlight

Xcode Cloud

iOS App

1.0 Prepare for Submission

Add Platform

General

App Information

App Review

History

App Store

TRUST & SAFETY

App Privacy

Ratings and Reviews

GROWTH & MARKETING

In-App Events

Custom Product Pages

Product Page Optimization

Promo Codes

Game Center

MONETIZATION

Pricing and Availability

In-App Purchases

Subscriptions

iOS App Version 1.0

Save

Add for Review

The assets and metadata below appear on your app's product page, when users install your app, and will be used for web engine search results once you release your app.

English (U.K.)

?

Previews and Screenshots

?

Adding accurate screenshots of your app on the newest devices can help you represent the app's user experience. Keep in mind that we'll use these screenshots for all display sizes and localizations. Screenshots are only required for iOS apps, and only the first 3 will be used on the app installation sheets.

iPhone

iPad

Apple Watch

View All Sizes in Media Manager

iPhone 6.9" Display

iPhone 6.5" Display

Drag up to 3 app previews and 10 screenshots here for iPhone 6.7" or 6.9" Displays.
(1320 x 2868px, 2868 x 1320px, 1290 x 2796px or 2796 x 1290px)

0 of 3 App Previews

0 of 10 Screenshots

Choose File

Delete All

Promotional Text

?

Main page for a new app in App Store Connect

Note that entering all the information is **long process** and you'll need to fill multiple pages:

8



iOS App

iOS App Version 1.0

Save

Add for Review

1.0 Prepare for Submission

[Add Platform](#)

The assets and metadata below appear on your app's product page, when users install your app, and will be used for web engine search results once you release your app.

English (U.K.) ?

General

App Information

App Review

History

Previews and Screenshots ?

Adding accurate screenshots of your app on the newest devices can help you represent the app's user experience. Keep in mind that we'll use these screenshots for all display sizes and localizations. Screenshots are only required for iOS apps, and only the first 3 will be used on the app installation sheets.

[iPhone](#)

[iPad](#)

[Apple Watch](#)

[View All Sizes in Media Manager](#)

App Store

TRUST & SAFETY

App Privacy

Ratings and Reviews

GROWTH & MARKETING

In-App Events

Custom Product Pages

Product Page Optimization

Promo Codes

Game Center

MONETIZATION

Pricing and Availability

In-App Purchases

Subscriptions



iPhone
6.9" Display



iPhone
6.5" Display

Drag up to 3 app previews and 10 screenshots here for iPhone 6.7" or 6.9" Displays.
(1320 x 2868px, 2868 x 1320px, 1290 x 2796px or 2796 x 1290px)

0 of 3 App Previews | 0 of 10 Screenshots | [Choose File](#) | [Delete All](#)

Promotional Text ?

Fill all the data in "App Information", "App Privacy", "Pricing and Availability", before adding your app for review

Here's what you'll need to provide and where to find it.

Sections to Complete

- **Main Page**
 - **Preview and Screenshots:** up to 3 app previews (optional), and up to 10 screenshots
 - **App Metadata:** description, keywords, support URL, marketing URL, version, copyright
 - **App Review Information:** Provide:
 - **Sign-in Information:** If your app requires login credentials for testing, create a test account and share them here.
 - **Contact Information:** Provide details for Apple to contact you during the review process.

- **Notes:** Add any additional context for the App Review team.
- **App Information**
 - **Localizable Information:** your app's name and subtitle
 - **General Information:** bundle ID, primary and secondary category, content rights information
 - **Age Rating:** a questionnaire with multiple selections and yes/no questions
 - **App Encryption Documentation:** specify whether your app uses non-exempt encryption (see tip below for most cases).
- **Pricing and Availability**
 - **Price Schedule:** starting price for your app (free or paid)
 - **App Availability:** countries where your app will be available to purchase or download
- **App Privacy**
 - **Privacy Policy:** a link to your app's Privacy Policy (required).
 - **Data Collection:** Specify your app's data collection and usage practices. This is a multi-step process where you'll need to disclose:
 - Types of data collected (e.g., location, identifiers).
 - How the data is used (e.g., analytics, advertising).
 - Whether the data is linked to user identity or used for tracking.

Tip: If your app doesn't use non-exempt encryption (this applies to most apps), add a `ITSAppUsesNonExemptEncryption` key and set the value to `NO` in your `Info.plist` file in Xcode. This will prevent extra encryption documentation steps when uploading your builds to App Store Connect. Read [this tip](#) for more details.

Before Submitting for Review

Make sure you've entered all the required app details. If any information is missing, you'll see an error when you click **Add for Review**:

Unable to Add for Review

The items below are required to start the review process:

- An unexpected error was encountered when submitting for review. If the issue persists please [contact us](#)
- You must select the level of frequency for each Apple content description in the Age Rating section.
- You must choose a build.
- You must complete the Contact Information section.
- You must set up Content Rights Information in [App Information](#).

[Show Details](#)

You have one or more errors on this page.

- English (U.K.) - Support URL - This field is required
- English (U.K.) - Keywords - This field is required
- English (U.K.) - Description - This field is required

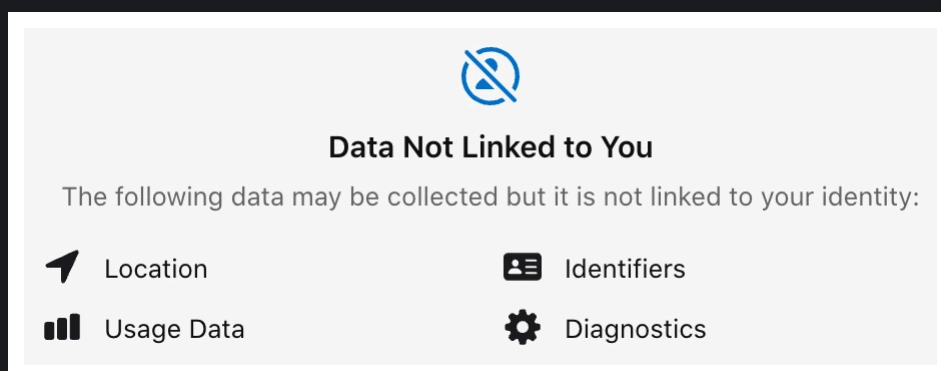
The assets and metadata below appear on your app's product page, when users install your app, and will be used for web engine search results once you release your app.

English (U.K.)   

Warning about required items before the app can be added for review

5. Create a Privacy Manifest in Xcode

When you specify your data collection practices in **App Store Connect**, Apple generates **privacy labels** for your app, which are displayed on your App Store listing. For example:



App Store privacy labels

Starting **November 12, 2024**, Apple also requires you to [add a Privacy Manifest to your app](#) before submitting it to App Store Connect.

What is a Privacy Manifest?

A **Privacy Manifest** is a file named `PrivacyInfo.xcprivacy` that documents your app's **data collection practices** and **API usage**. This file helps Apple understand what data your app collects, how it's used, and why.

How to Create a Privacy Manifest in Xcode

Follow these steps to create a Privacy Manifest for your app:

- Open the `ios/Runner.xcworkspace` project in Xcode.
- Right-click the **Runner** folder and select **New File from Template...**
- Scroll down to the **Resource** section, choose **App Privacy**, and click **Next**.
- Create the file with the default name `PrivacyInfo`.

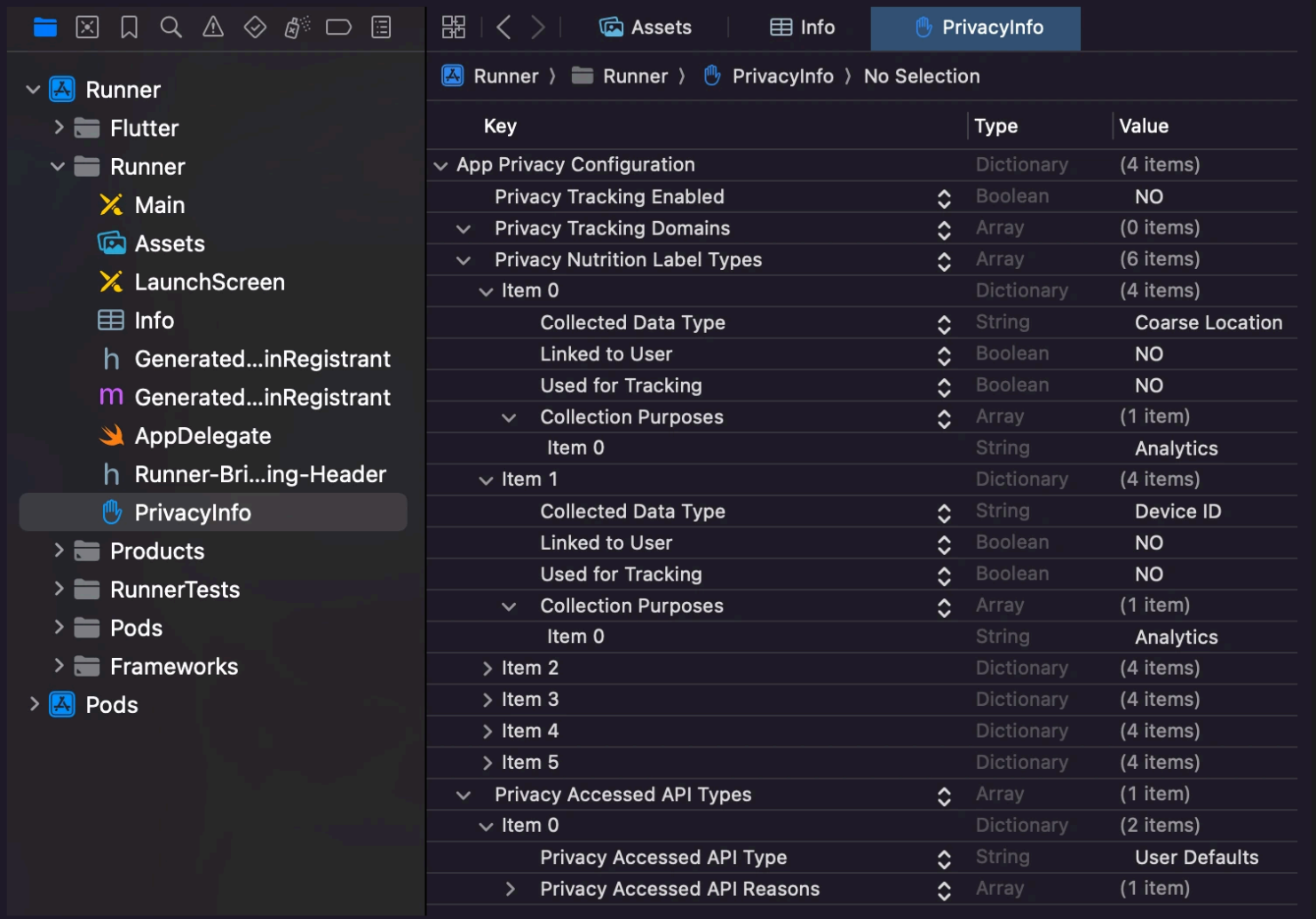
What to Add to the Privacy Manifest

Once the file is created, add the following values to the **App Privacy Configuration** section:

- **NSPrivacyTracking**: Indicates whether your app uses tracking, as defined by Apple's App Tracking Transparency framework.
- **NSPrivacyTrackingDomains**: Lists internet domains your app uses for tracking purposes.
- **NSPrivacyCollectedDataTypes**: Details the types of data collected by your app (e.g., location, identifiers).
- **NSPrivacyAccessedAPITypes**: Describes the APIs your app accesses and the reasons for their use (e.g., **User Defaults** for local storage).

These should match the data collection types you declared in App Store Connect.

Once all the information is complete, your `PrivacyInfo` file should look similar to this:



Xcode privacy manifest file

Example File

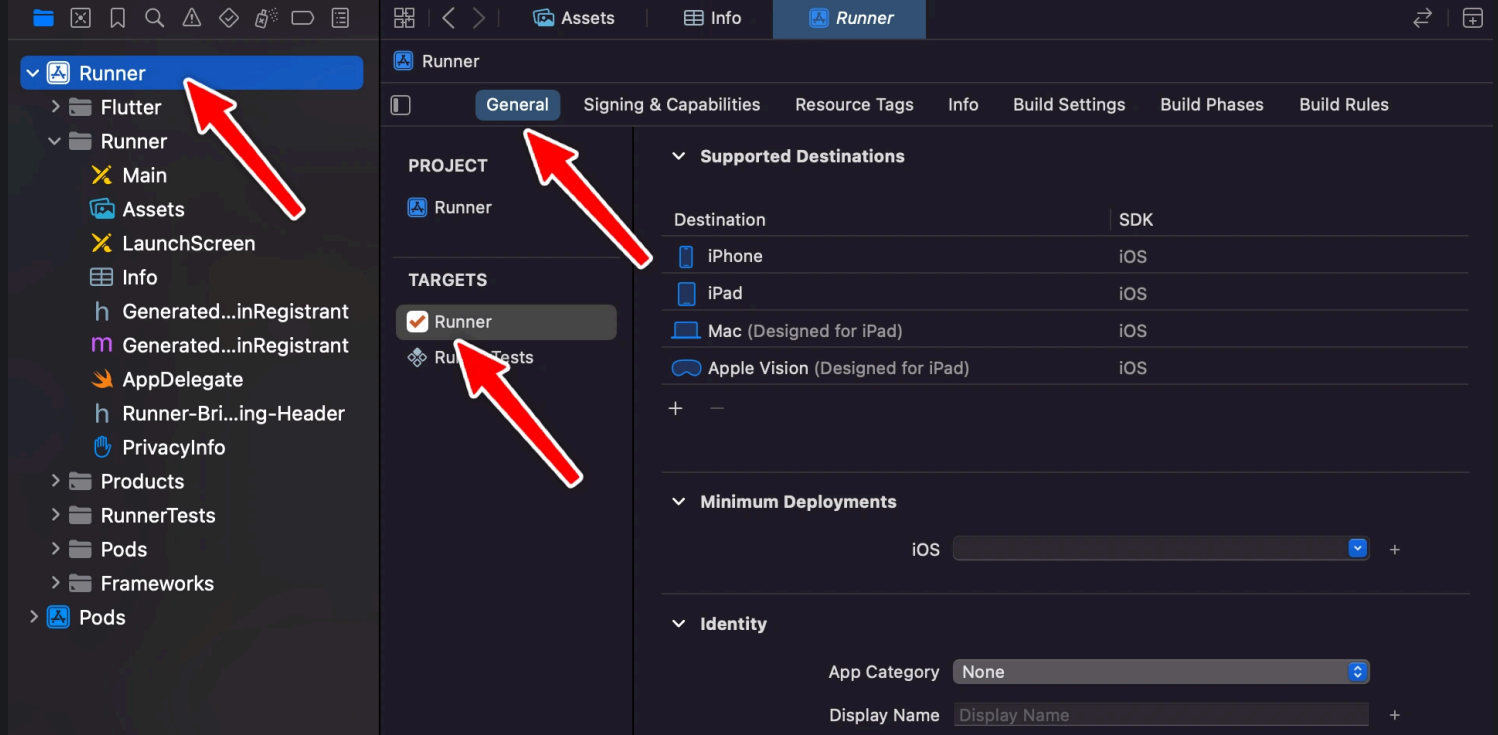
Here's an example of the completed `PrivacyInfo.xcprivacy` for one of my apps:

- [Example PrivacyInfo.xcprivacy](#)

To learn more, read the official Apple docs about [User privacy and data use](#) and [Privacy manifest files](#).

6. Update the Xcode project settings

Before building and submitting your app to **App Store Connect**, review your **Xcode project settings** to ensure everything is configured correctly. You can find these settings in **Runner > General**:



Xcode project settings (General tab)

Key Xcode project settings

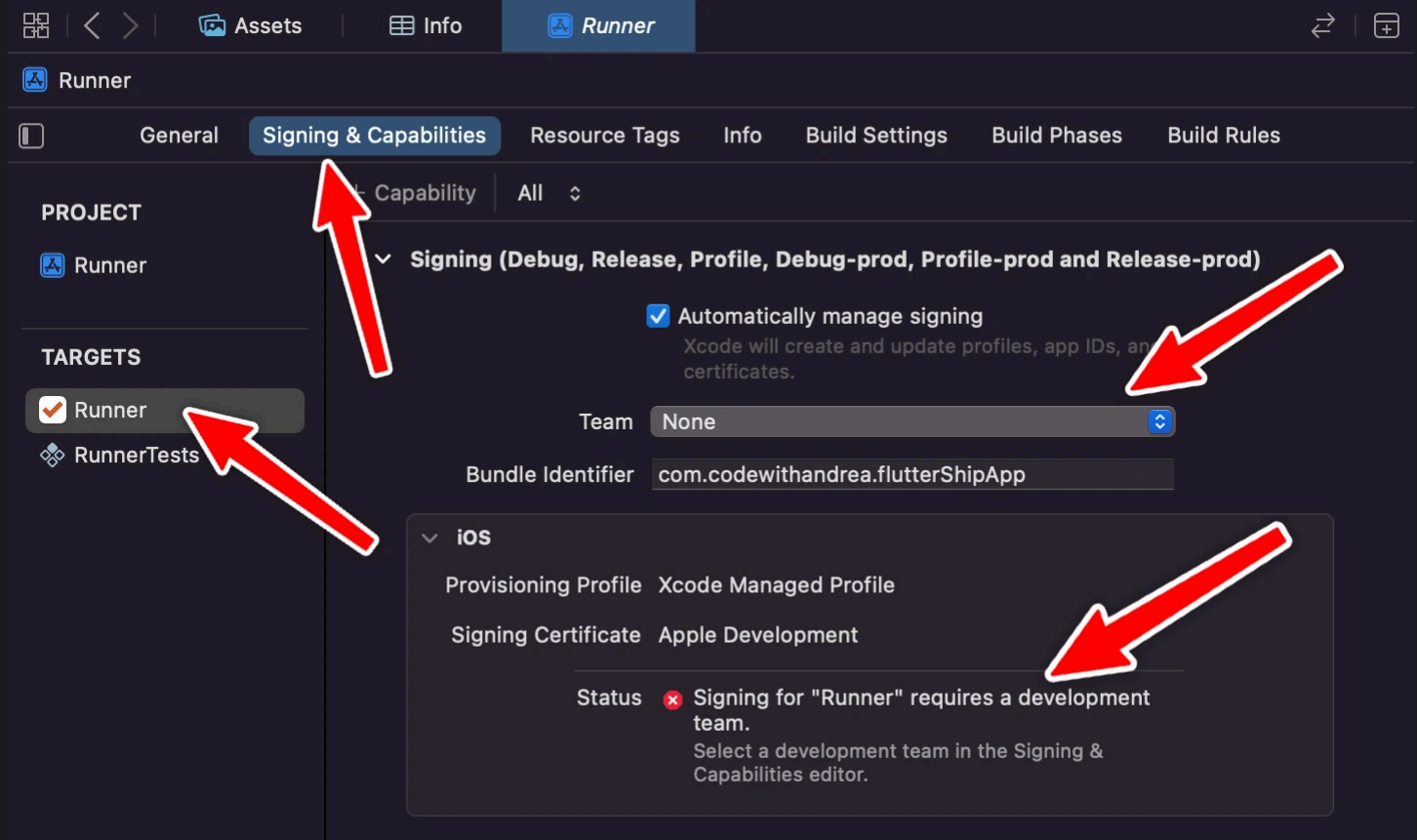
Here are the key settings you need to review and update:

- **Supported Destinations:** In addition to **iPhone**, decide if your app should support **iPad**, **Mac**, and **Apple Vision**
- **Minimum Deployment Version:** Targeting an iOS version that is **2 years old** ensures your app will be compatible with **over 90% of devices** ([source](#)).
- **Identity:** Double-check the **App Category**, as it impacts your app's discoverability on the App Store. Other fields like **Display Name**, **Bundle Identifier**, **Version**, and **Build Number** are derived from your `pubspec.yaml` file, so there should be no need to change them.
- **Deployment Info:** Review and update the **supported orientations** based on your app's design.
- **App Icons and Launch Screen:** Verify that they look as intended.

Tip: use packages like [flutter_launcher_icons](#) and [flutter_native_splash](#) to generate consistent app icons and launch screens for all platforms.

Code Signing

In addition to the general settings, you'll need to review the **Signing & Capabilities** tab:



Signing & Capabilities tab in Xcode

- **Enable Automatic Signing:** Ensure **Automatically manage signing** is enabled.
- **Select Your Team:** In the dropdown, choose your **Team**. This allows Xcode to automatically generate the necessary **certificate** and **provisioning profile** (if they don't already exist).

By default, a **development** certificate is created (rather than a **distribution** one). This is normal and won't prevent you from uploading your app to App Store Connect. You can find all your certificates in the [Apple Developer Portal](#).

App Version and Build Number

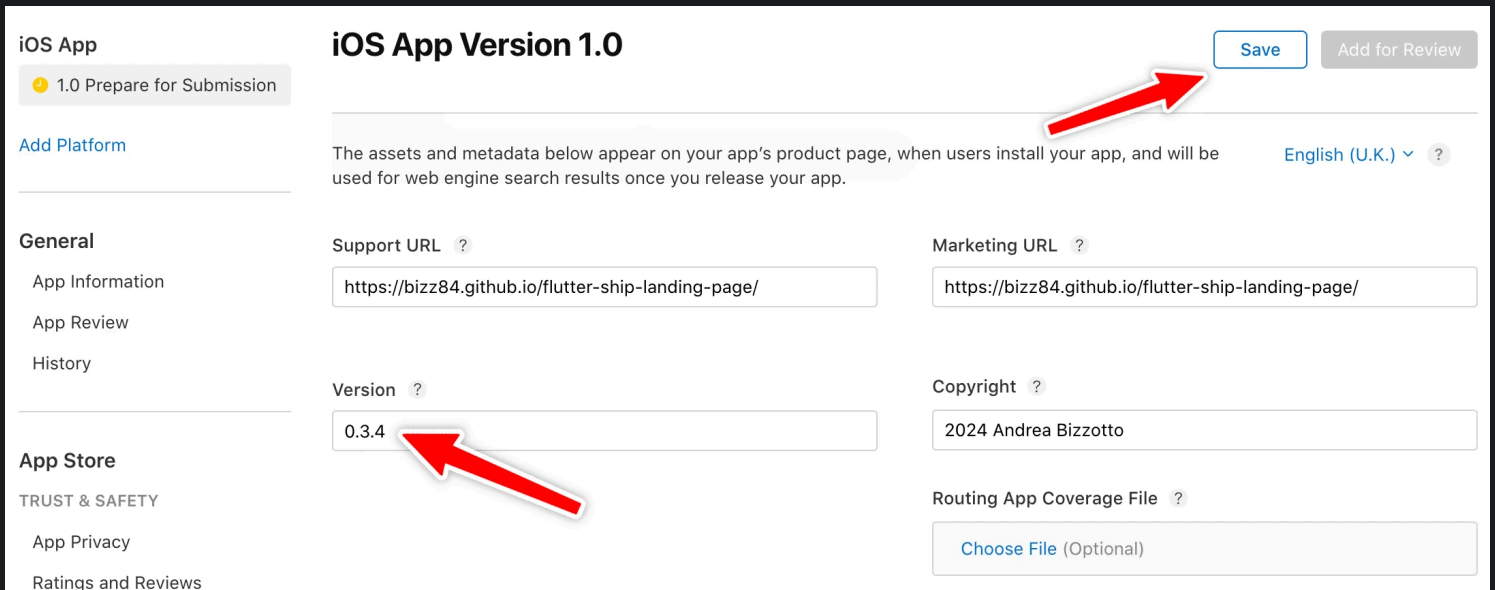
In Flutter, the **version** and **build number** are defined in the `pubspec.yaml` file under the `version` key:

```
version: 0.3.4+18
```

In **App Store Connect**, you'll need to manually set the app version number to match the version in your `pubspec.yaml`. Here's how:

1. Go to **App Store Connect > My Apps > Your App**.

2. Scroll down to the metadata and set the correct **version number** to match what's in your `pubspec.yaml`:



The screenshot shows the 'iOS App Version 1.0' configuration page in App Store Connect. The left sidebar includes sections for 'iOS App' (with a '1.0 Prepare for Submission' status), 'Add Platform', 'General' (with sub-items: App Information, App Review, History), 'App Store', 'TRUST & SAFETY' (with sub-items: App Privacy, Ratings and Reviews), and 'App Store Connect'. The main content area displays various metadata fields: 'Support URL' and 'Marketing URL' both point to 'https://bizz84.github.io/flutter-ship-landing-page/'; 'Version' is set to '0.3.4' (highlighted with a red arrow); 'Copyright' is '2024 Andrea Bizzotto'; and 'Routing App Coverage File' has a 'Choose File (Optional)' button. A 'Save' button (highlighted with a red arrow) and an 'Add for Review' button are at the top right. A language selector shows 'English (U.K.)'.

Setting the version number in App Store Connect

7. Build, upload, and submit your app to App Store Connect

Once you've verified your **Xcode settings** and **versioning**, it's time to build your app bundle and upload it to **App Store Connect** for review.

Building the iOS App Bundle

To build the iOS app bundle, run the following command:

```
flutter build ipa
```

Depending on your app's configuration, you might need to pass additional flags. To see all supported arguments, run: `flutter build ipa --help`.

If your app supports multiple flavors, specify the flavor using the `--flavor` flag. For example:

```
# To build the prod flavor
flutter build ipa --flavor prod -t lib/main_prod.dart --dart-define-from-file=.env.prod
```

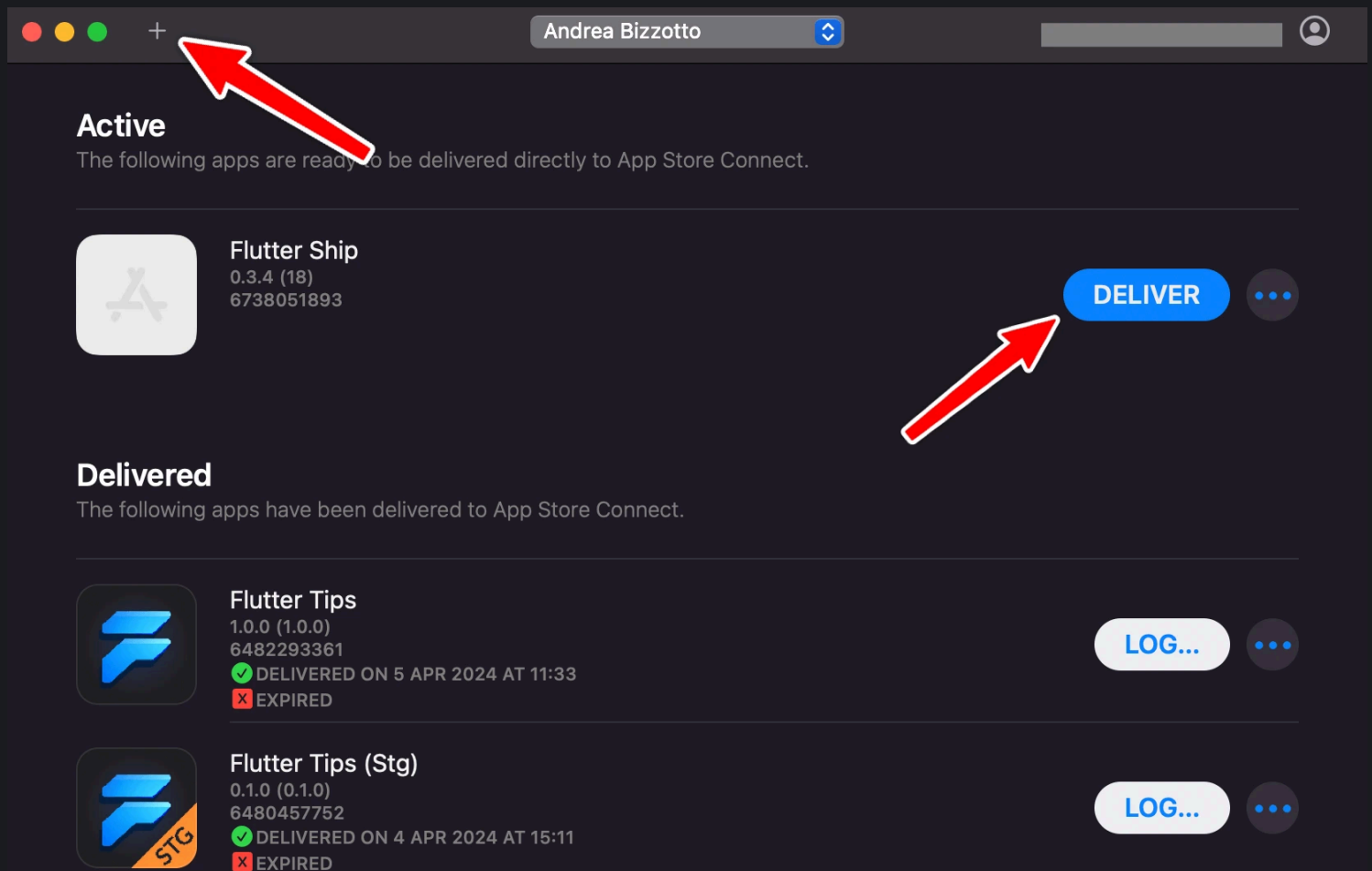
This command will generate an **IPA** file (iOS App Archive) in the `build/ios/ipa/` directory. The IPA is the format used to distribute iOS apps.

Uploading the App Bundle to App Store Connect

There are multiple ways to upload your app bundle, but for first-time uploads, I recommend using the Apple Transporter app. Transporter simplifies the upload process by handling both the upload and validation for you.

How to Use Transporter:

- Open the Apple Transporter app and sign in with your Apple ID.
- Drag and drop the `build/ios/ipa/*.ipa` file into the app.
- Click **Deliver** to upload your app bundle:



After a few minutes, Transporter will show the app as **Ready for Internal Testing**:

Delivered

The following apps have been delivered to App Store Connect.



Flutter Ship

0.3.4 (18)

6738051893

✓ DELIVERED ON 14 NOV 2024 AT 12:12

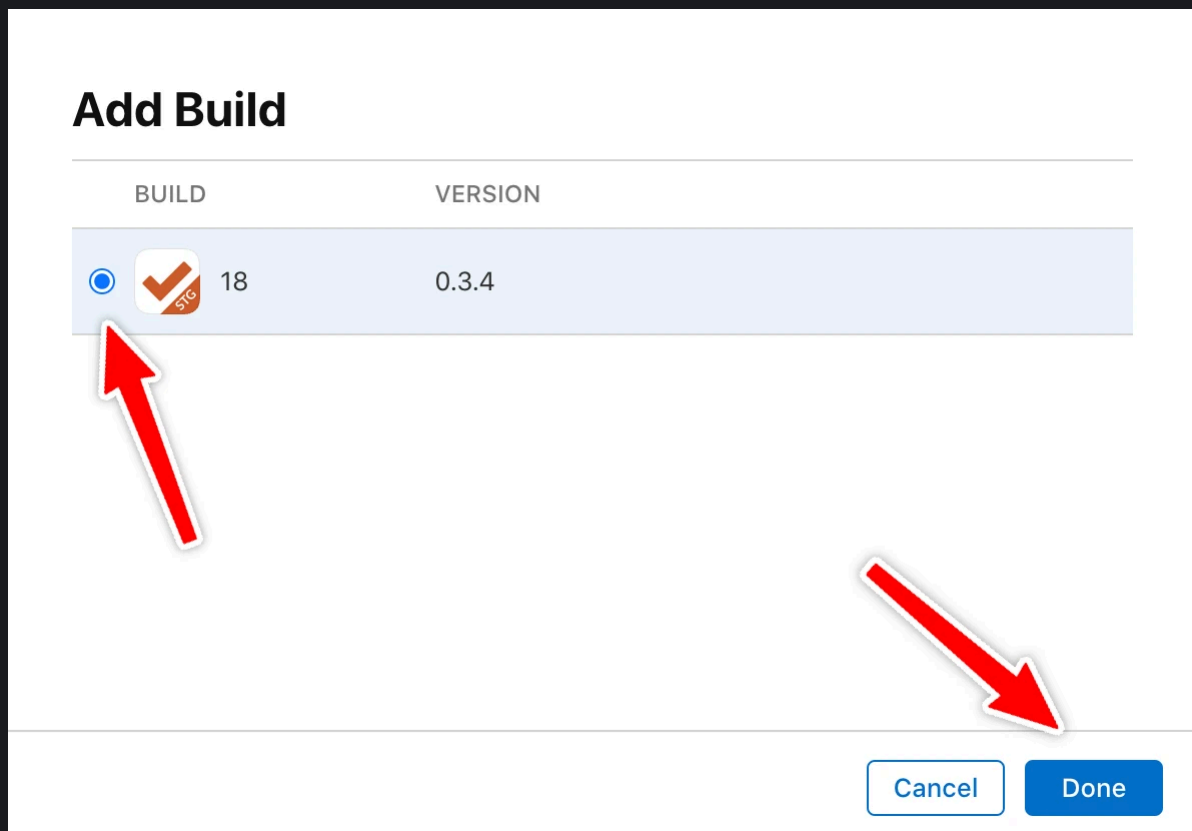
⌚ THE APP IS READY FOR INTERNAL TESTING

LOG...



Submitting Your App for Review

Go to your app in **App Store Connect**, scroll down to **Build**, click the + icon, select your build, and click **Done**:



Then, click **Add for Review** at the top of the page, then **Submit to App Review**:

iOS App Version 0.3.4

✓ Saved

Add for Review

Build

BUILD	VERSION	HAS APP CLIP
 18	0.3.4	NO



18

0.3.4

NO

Included Assets



App Icon

What Happens Next?

Your app will now enter Apple's **review queue**. Apple's review team will assess your app for compliance with their guidelines. If everything checks out, you'll receive a notification, and your app will be ready for release!

Wrapping Up

Congratulations! You've successfully submitted your iOS app to **App Store Connect**—the final step before Apple reviews it for release on the App Store. 🎉

What's Next?

While waiting for App Review, it's a good idea to get familiar with Apple's review guidelines and resources to ensure a smooth approval process. Start with the [App Review](#) page, which includes helpful documentation like:

- [App Review Guidelines](#)
- [Human Interface Guidelines](#)
- [Guidelines for Using Apple Trademarks and Copyrights](#)
- [Submitting for review](#)
- [Avoiding common issues](#)

Submitting App Updates to App Store Connect

Once your app is approved, you'll likely submit updates to add new features and fix bugs. While using the **Transporter app** works for occasional uploads, it can quickly become tedious if you update your app frequently.

A better solution? **Automate the process** with a build script that uses the `xcrun` command to automatically build and upload your app to App Store Connect.

To learn more about this, read:

- [iOS App Store: Build and Upload Script](#)

Scaling with CI/CD Pipelines

If you work as part of a team or release updates often, consider setting up **CI/CD pipelines** to automate your app builds, testing, and releases. CI/CD enables you to:

- Save time by automating repetitive tasks.
- Run specific workflows (e.g. run tests) when certain events are triggered (e.g. when pushing a branch).
- Distribute apps more efficiently via TestFlight or the App Store.

To dive deeper into these topics, and many more, you can explore my new course. 🙌

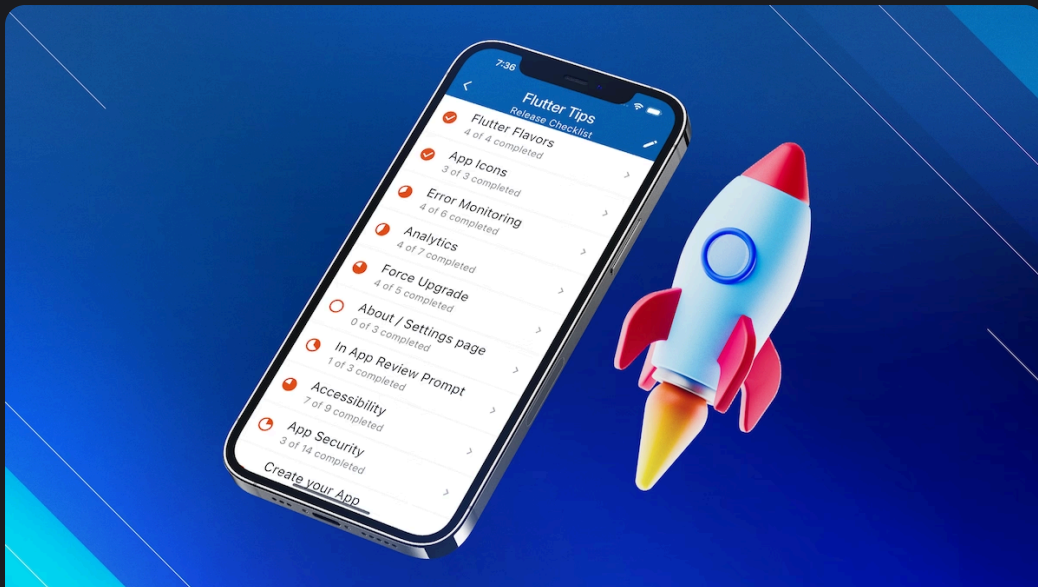
New Course: Flutter in Production

When it comes to **shipping** and **maintaining** apps in production, there are many important aspects to consider:

- **Preparing for release:** splash screens, flavors, environments, error reporting, analytics, force update, privacy, T&Cs.
- **App Submissions:** app store metadata & screenshots, compliance, testing vs distribution tracks, dealing with rejections.
- **Release automation:** CI workflows, environment variables, custom build steps, code signing, uploading to the stores.
- **Post-release:** error monitoring, bug fixes, addressing user feedback, over-the-air updates, feature flags & A/B testing.

My latest course will help you get your app to the stores faster and with fewer headaches.

If you're interested, you can learn more and enroll here. 📌



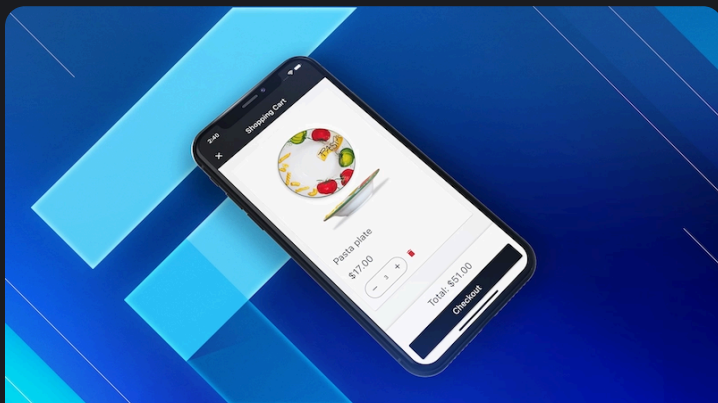
Flutter In Production

INTERMEDIATE TO ADVANCED

Learn about flavors, environments, error monitoring, analytics, release management, CI/CD, and finally ship your Flutter apps to the stores. 🚀

Want More?

Invest in yourself with my high-quality Flutter courses.



Flutter Foundations Course

INTERMEDIATE TO ADVANCED

Learn about State Management, App Architecture, Navigation, Testing, and much



Flutter & Firebase Masterclass

INTERMEDIATE TO ADVANCED

21

more by building a Flutter eCommerce app on iOS, Android, and web.

Learn about Firebase Auth, Cloud Firestore, Cloud Functions, Stripe payments, and much more by building a full-stack eCommerce app



The Complete Dart Developer Guide

BEGINNER

Learn Dart Programming in depth. Includes: basic to advanced topics, exercises, and projects. Last updated to Dart 2.15.



Flutter Animations Masterclass

INTERMEDIATE

Master Flutter animations and build a completely custom habit tracking application.



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